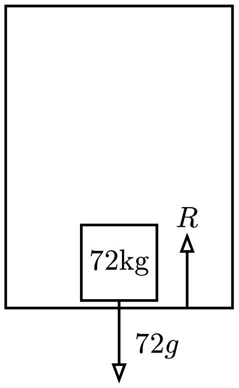
Part A

1. i.



ii. Then $a = 0$ so the normal reaction is $-72g\text{N} = 710\text{N}$

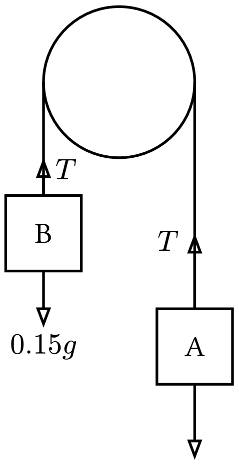
iii. Then $a = 2$ so $-72g + R = 72(2) \therefore R = 710 + 144 = 854\text{N}$

iv. Then $a = -3$ so $-72g + R = 72(-3) \therefore R = 710 - 216 = 494\text{N}$

v. Then $a = -2$ so $-72g + R = 72(-2) \therefore R = 710 - 144 = 570\text{N}$

vi. Then $a = 3$ so $-72g + R = 72(3) \therefore R = 710 + 216 = 926\text{N}$

2. i.



ii.
$$\begin{cases} 0.25g - T = 0.25a \\ T - 0.15g = 0.15a \end{cases}$$

iii.
$$\begin{aligned} \therefore 0.4a &= 0.1g \\ a &= 2.45\text{ms}^{-2} \end{aligned}$$

iv. $T - 0.15g = 0.15(2.45)$
$$\begin{aligned} T &= 0.15 \times 2.45 + 0.15 \times 9.8 \\ &= 1.8375 \approx 1.84 \end{aligned}$$

v.
$$\begin{aligned} 1 &= ut + \frac{1}{2}at^2 \\ &= 0 + \frac{1}{2} \times 2.45 \times t^2 \end{aligned}$$

$t = \sqrt{\frac{2}{2.45}} = 0.904\text{s}$

3.
$$\begin{cases} T - 3g = 3a \\ 9g - T = 9a \end{cases}$$

$$\therefore 6g = 12a \Rightarrow a = 4.9\text{ms}^{-2}$$

$$\text{so } T - 3g = 3(4.9) \therefore T = 44.1\text{N}$$

4. i. Let T_1 be the tractive force of the engine. Then

$$\begin{aligned} T_1 - 80(60000) - 50(12000) &= (60000 + 12000)a \\ \text{speed} &= \text{constant} \therefore a = 0 \\ T_1 - 80(60000) - 50(12000) &= 0 \Rightarrow T_1 = 5400 \text{ kN} \end{aligned}$$

ii. Let T_2 be the tension in the coupling between engine and truck. Then

$$\begin{aligned} T_2 - 50(12000) &= 0 \text{ kN } (a = 0) \\ \text{so } T_2 &= 600 \text{ kN} \end{aligned}$$

5. i. Let T_1 be the tension in the string. Then

$$\begin{cases} T = 4a \\ 1.5g - T = 1.5a \end{cases}$$

$$\therefore a = \frac{1.5g}{5.5} = \frac{3g}{11} = 2.67\text{ms}^{-2}$$

$$\begin{aligned} T &= 4a \\ &= 4(2.67) = 10.7\text{N} \end{aligned}$$

6. i. On the 3 kg cube:

$$\begin{aligned} R_1 &:= \text{reaction force between 3kg cube and 7kg cube} \\ 3g - R_1 &= 0 \qquad (a = 0) \\ \therefore R_1 &= 29.4\text{N} \end{aligned}$$

ii. On the 7 kg cube:

$$\begin{aligned} R_2 &:= \text{reaction force between 7kg cube and table} \\ (3 + 7)g - R_2 &= 0 \qquad (a = 0) \\ \therefore R_2 &= 10g \\ &= 98\text{N} \end{aligned}$$

Part B

1. i. Let T be the tension in the string. Then

$$\begin{cases} T = m_1 a \\ m_2 g - T = m_2 a \end{cases}$$

then $m_2 g = m_1 a + m_2 a = a(m_1 + m_2)$

$$\therefore a = \frac{m_2 g}{m_1 + m_2}$$

ii. We know that $s = ut + \frac{1}{2}at^2$

Then

$$\begin{aligned} d &= 0 + \frac{1}{2}at^2 \\ t &= \sqrt{\frac{2d}{a}} \\ &= \sqrt{\frac{\frac{2d}{m_2 g}}{m_1 + m_2}} \\ &= \sqrt{\frac{2d(m_1 + m_2)}{m_2 g}} \end{aligned}$$

2. i.

$$\begin{cases} T - 8.5g = 8.5a \\ 14.5g - T = 14.5a \end{cases}$$

$$\therefore 6g = 23a; a = 2.56\text{ms}^{-2}.$$

For particle A,

$$v^2 = u^2 + 2as$$

When $u = 0$, $v = \sqrt{2as}$

$$v_A = \sqrt{2 \times 2.56 \times 3.5} = 4.23\text{ms}^{-1}$$

Then when B is 7m above the ground, its velocity is -4.23ms^{-1}

ii. We can say that the displacement relative to the point where B is 7m above the ground is given by the equation

$$s = ut + \frac{1}{2}at^2$$

$u = -4.23, a = 9.8.$

When B is 7m above the ground, let $s = 0$. Then

$$\begin{aligned} ut + \frac{1}{2}at^2 &= 0 \\ (-4.23)t + 4.9t^2 &= 0 \\ t(-4.23 + 4.9t) &= 0 \end{aligned}$$

So $t = 0$s or $t = 0.8632$s.

Therefore, midway between these times, particle B is at its max height.

Let $t = 0.4316$s. Then

$$\begin{aligned} s &= ut + \frac{1}{2}at^2 \\ &= (-4.23)(0.4316) + \frac{1}{2}(9.81)(0.4316)^2 \\ &= -0.9119724768\text{m} \end{aligned}$$

Therefore the max distance that B reaches above the ground is $7\text{m} + 0.9129\text{m} = 7.91\text{m}$.

3. i.

$$\begin{cases} T - 4g = 4a \\ 10g - T = 10a \end{cases}$$

Therefore $6g = 14a \Rightarrow a = 4.3\text{ms}^{-2}$

We can say that

$$s = ut + \frac{1}{2}at^2$$

For particle B,

$$\begin{aligned} s_B &= 0 + \frac{1}{2} \times 4.3 \times (0.2)^2 \\ &= 0.086\text{m} \end{aligned}$$

Then at $t = 0.2$ particle B's displacement is -0.914m and its acceleration is 9.8ms^{-1}.

Also, at 0.2s

$$\begin{aligned} v &= \sqrt{u^2 + 2as} \\ &= \sqrt{0^2 + 2 \times 4.3 \times 0.084} \\ &= 0.8617192118\text{ms}^{-1} \end{aligned}$$

ii. Then we can say that at $t = 0.2$, particle B has 0.914m to fall and has a velocity of $0.8617192118\text{ms}^{-1}$ downwards.

Then, we can say that

$$\begin{aligned} s &= ut + \frac{1}{2}at^2 \\ 0.914 &= 0.8617192118t + 4.9t^2 \\ t = 0.35282 &\text{ or } t = -0.52868 \\ \text{but } t > 0 &\text{ so } t = 0.35282. \end{aligned}$$

$\boxed{t = 0.353\text{s}}$